DHANUSH V C

Aspiring AI & Machine Learning Engineer
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PROFESSIONAL SUMMARY

Aspiring AI/ML engineer with hands-on experience building end-to-end machine learning pipelines, computer vision systems, and biomedical signal-processing models. Skilled in Python, PyTorch, and deep learning frameworks, with experience translating research problems into working technical solutions across healthcare and industrial vision applications. Strong foundation in data preprocessing, model evaluation, and applied ML experimentation.

TECHNICAL SKILLS

Programming Languages: Python
Machine Learning: Deep Learning, NLP, Regression, Model Evaluation, Computer Vision
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV
Computer Vision Tools: YOLO, SSD, OCR (Tesseract, EasyOCR)
Backend & APIs: Flask, Django, REST APIs
Databases: MySQL, MongoDB
Developer Tools: Git, GitHub

WORK EXPERIENCE

Data Analyst Intern — Zidio *Mar 2025 – Jun 2025*
Bangalore, India

- Completed the Zidio AI & Data Science internship, applying machine learning techniques to real-world datasets
- Performed data preprocessing and feature engineering to prepare datasets for model training
- Built end-to-end ML pipelines in Python, covering data ingestion through model output
- Evaluated model performance using standard metrics and structured experimentation

PROJECTS

Robust Multimodal Non-Invasive Disease Prediction Using Synthetic Spectral Data *Nov 2025 – Present*

- Built a synthetic near-infrared (NIR) pipeline for biomedical signal generation
- Modeled noise, degradation, and signal variability to simulate real-world conditions
- Combined spectral and clinical data for machine learning-based regression
- Benchmarked Random Forest, CNN, and Transformer models for prediction accuracy
- Validated performance using RMSE, MAE, R^2, and k-fold cross-validation

Robust Object Detection Under Extreme Environmental Degradation *Mar 2025 – Jul 2025*

- Built an image degradation engine simulating fog, blur, and noise conditions
- Simulated real-world visual distortions to stress-test detection models
- Benchmarked YOLO and SSD models using mAP and precision-recall curves
- Improved model robustness through data augmentation and adversarial training
- Analyzed latency and edge-deployment trade-offs for real-time use cases

Cross-Domain Adaptation Framework for Small-Scale Industrial Vision Systems *Oct 2024 – Feb 2025*

- Built a defect detection system for limited labeled-data scenarios
- Identified and analyzed cross-domain performance degradation
- Diagnosed distribution shift as the root cause of model failures
- Applied transfer learning and domain adaptation techniques to close the gap
- Validated results using performance metrics, ablation studies, and robustness tests

PUBLICATIONS

Non-Invasive Continuous Blood Glucose Monitoring Using Near-Infrared Spectroscopy and Machine Learning *2025*

ICECIT 2025, IEEE — Author: Dhanush V C

- Built a synthetic NIR pipeline for glucose signal simulation
- Simulated noise and signal distortion conditions
- Developed Random Forest, CNN, and Transformer regression models
- Compared models for non-invasive glucose prediction accuracy
- Validated results using RMSE, MAE, R^2, and k-fold cross-validation

EDUCATION

Bachelor of Engineering — PES Institute of Technology and Management *2022 – 2026*

Bangalore, India | Field of Study: Engineering

- Built strong problem-solving and debugging skills through hands-on engineering projects

Class XII — St. Charles Composite Pre-University College *2020 – 2022*

Bangalore, India | Field of Study: Science

- Built strong fundamentals in mathematics, physics, and programming

VOLUNTEERING

IEEE Technical Event Volunteer *Apr 2025*

Bangalore, India

- Collaborated with the IEEE team on technical event execution
- Coordinated participants across multiple sessions
- Managed on-ground operations and event flow
- Facilitated technical discussions with attendees and industry experts